



AALBORG
UNIVERSITY HOSPITAL

Bakteriel meningitis

5. semester, bachelor

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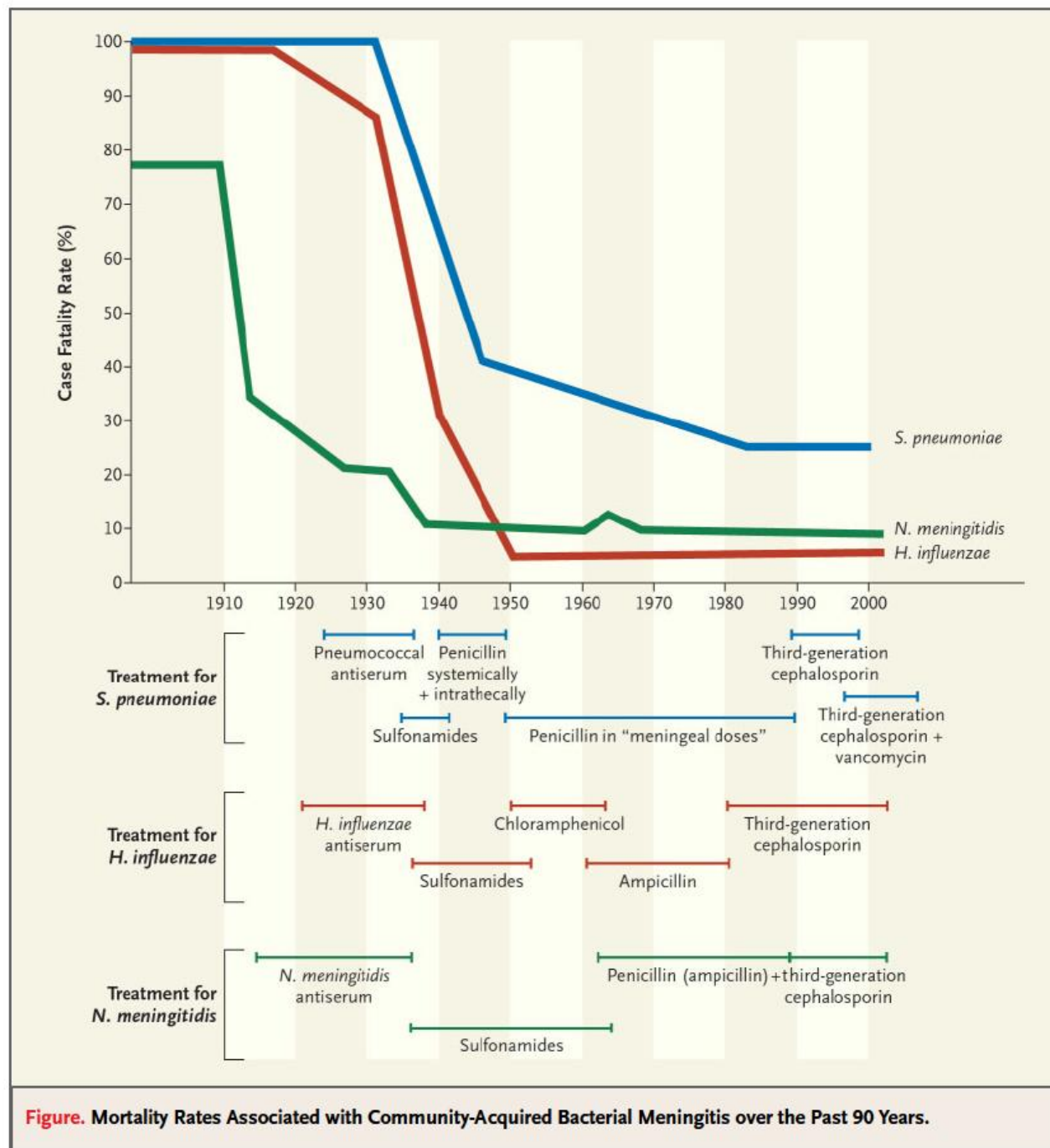
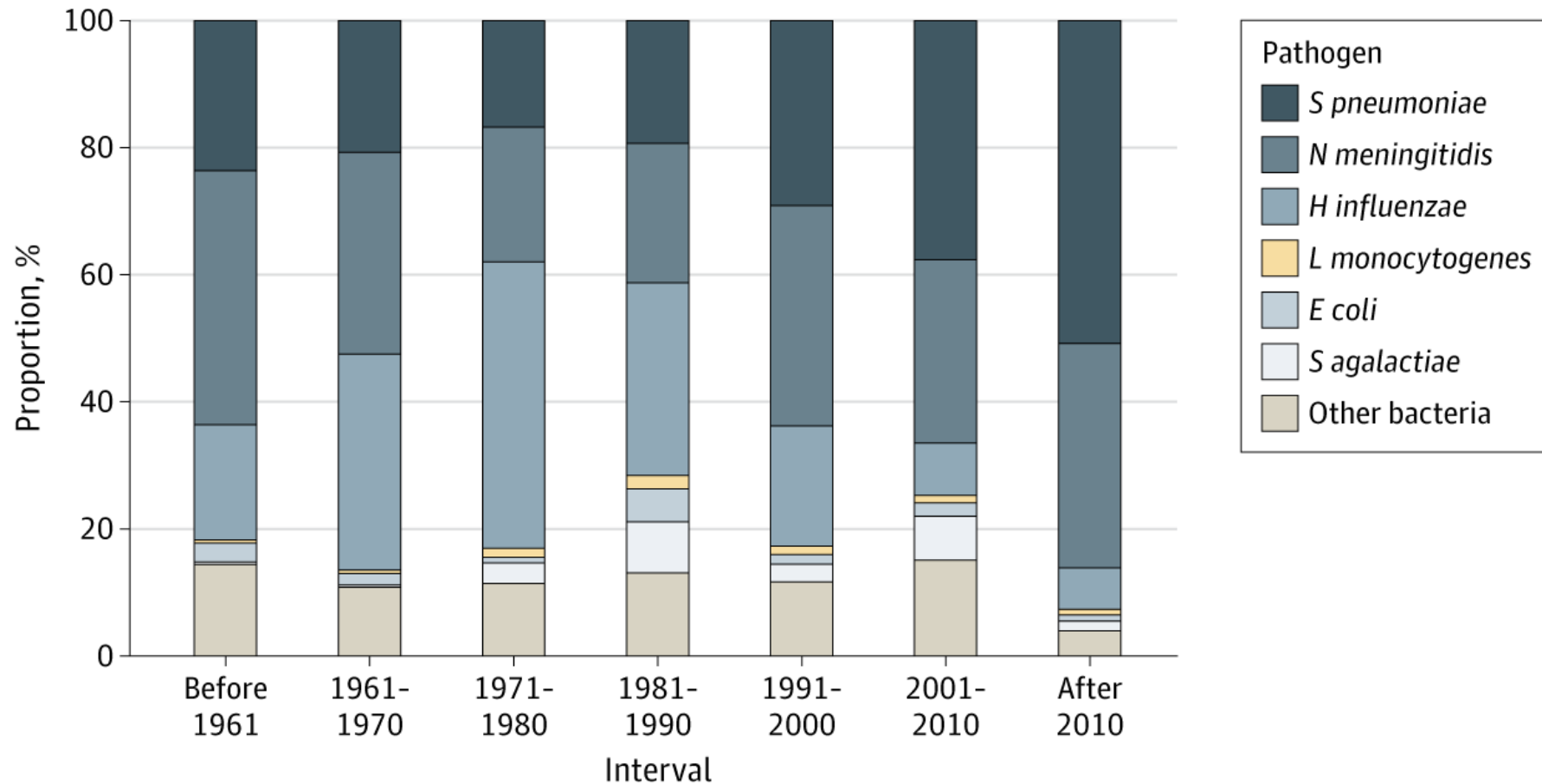


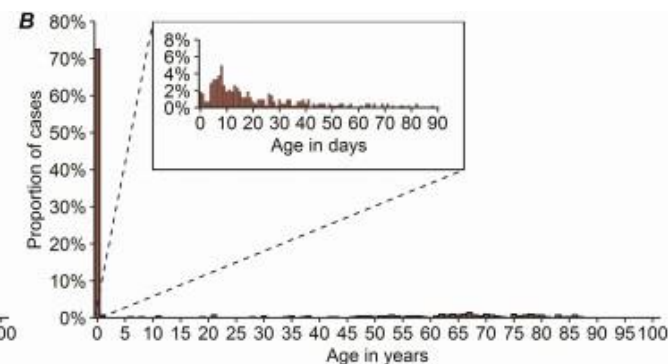
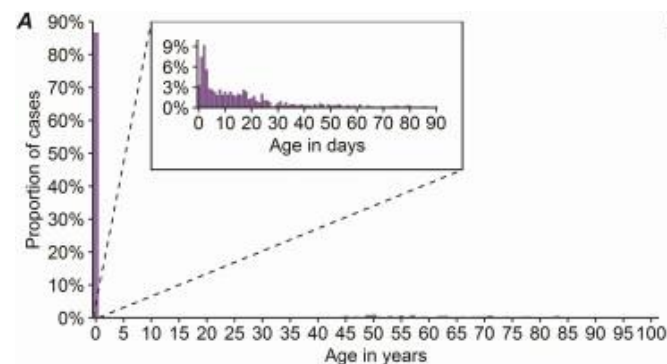
Figure. Mortality Rates Associated with Community-Acquired Bacterial Meningitis over the Past 90 Years.

Epidemiology – overall



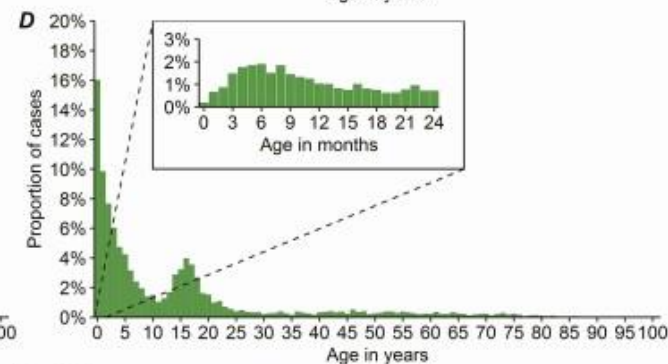
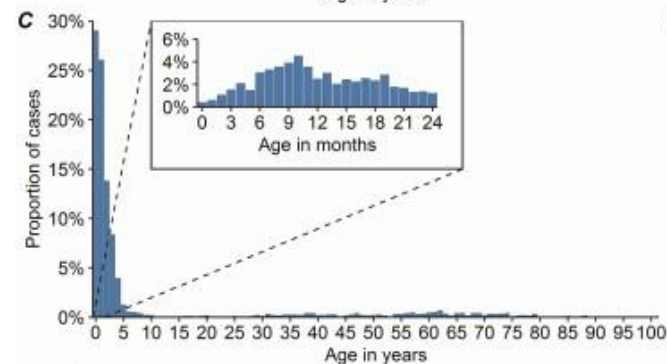
Aldersgrupper

GBS



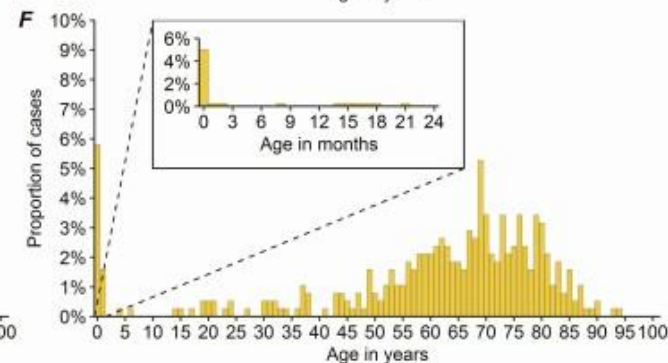
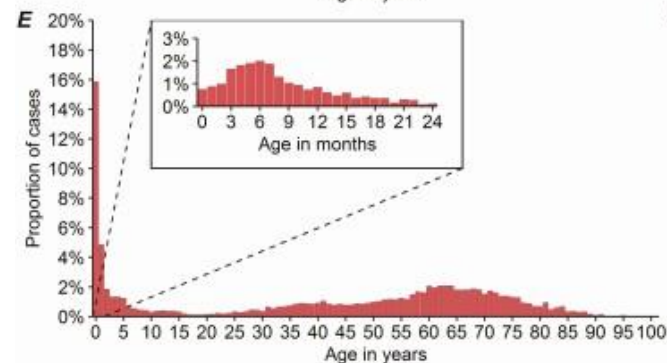
E. coli

Hib



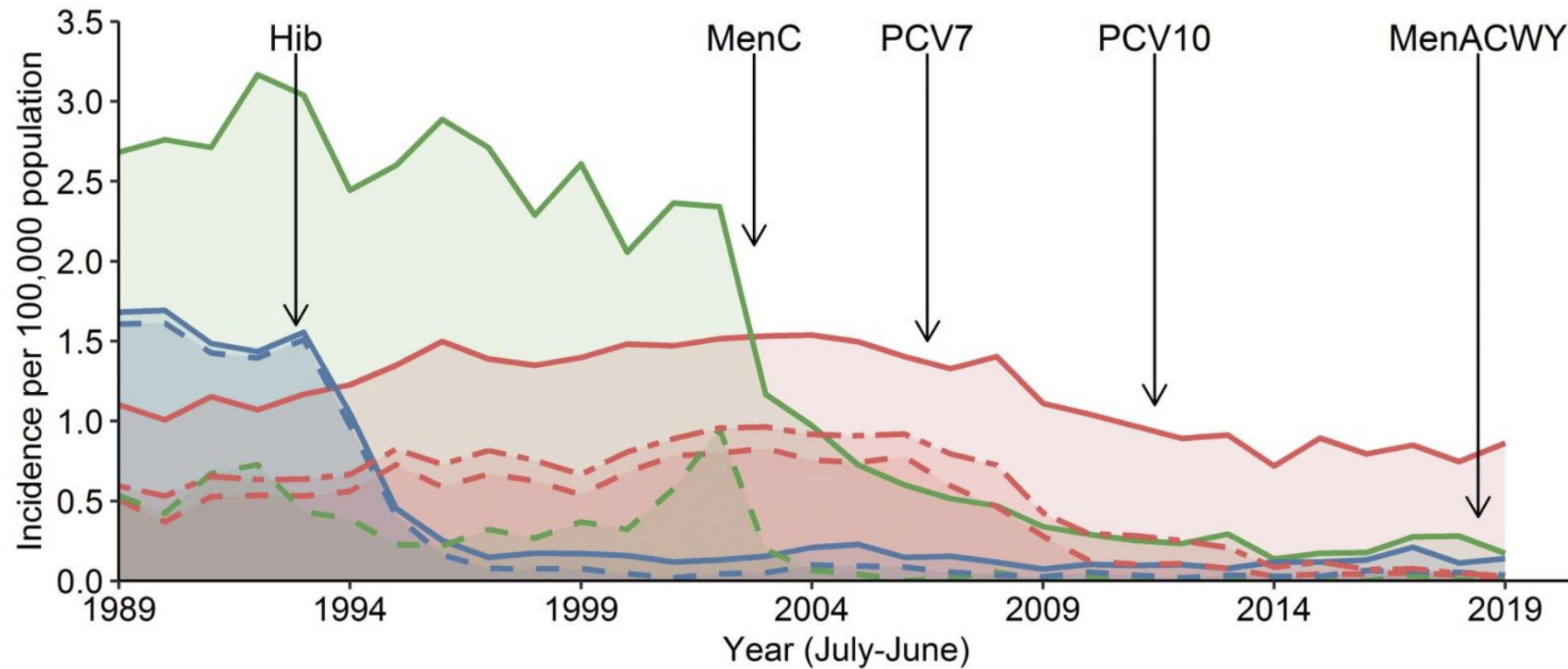
Meningokokker

Pneumokokker

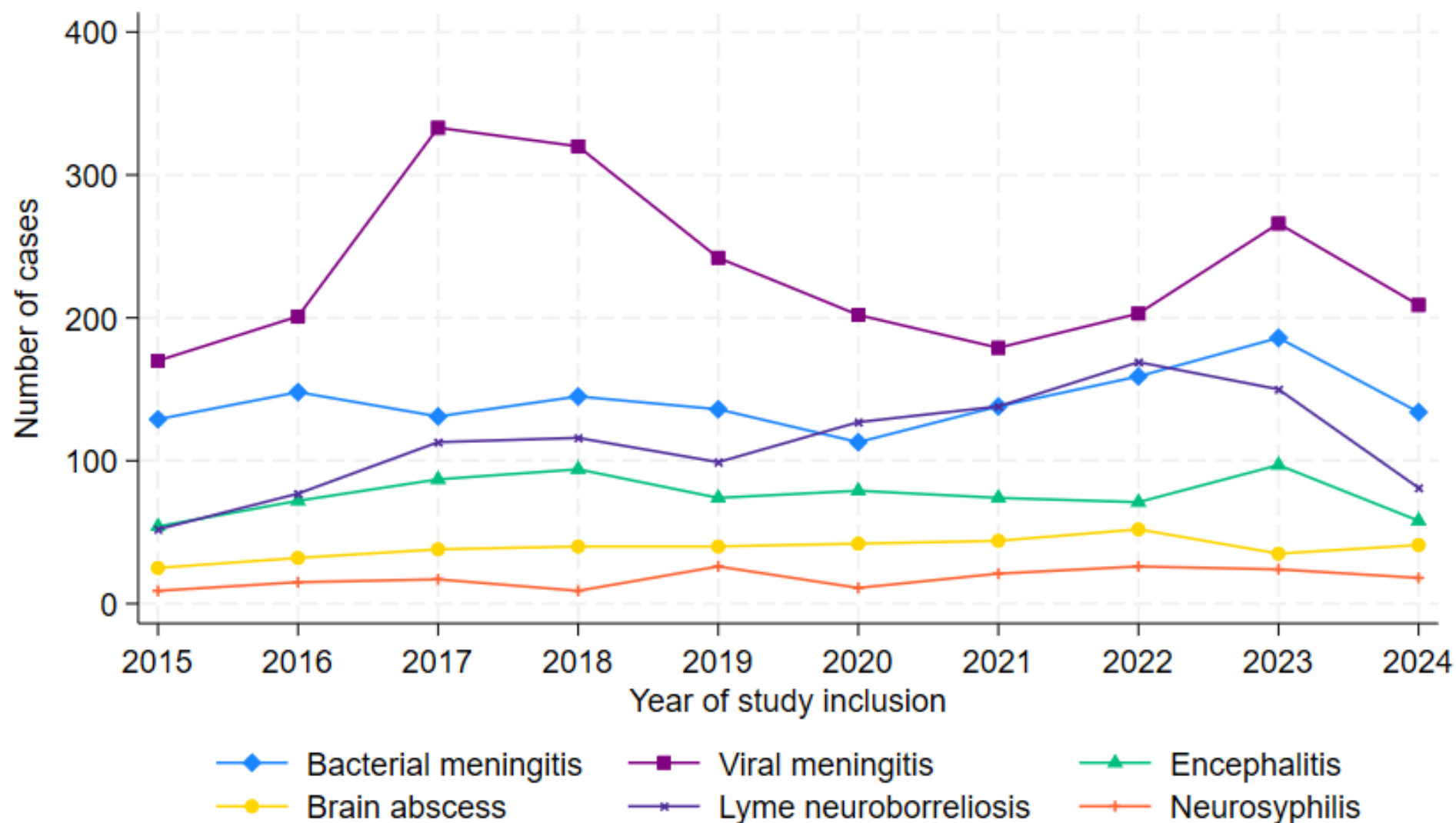


Listeria

Epidemiology – the Netherlands



CNS infektioner blandt voksne i Dk



Bakteriel meningitis: DASGIB 2015-2025

Patogen	N (%)
<i>S. pneumoniae</i>	562 (35)
<i>S. aureus</i>	116 (7)
<i>L. monocytogenes</i>	100 (6)
<i>N. meningitidis</i>	79 (5)
<i>H. influenzae</i>	71 (4)
<i>E. coli</i>	34 (2)
Others (incl. BHS)	309 (19)
Unidentified	336 (21)
Total	1,607



Patogenese

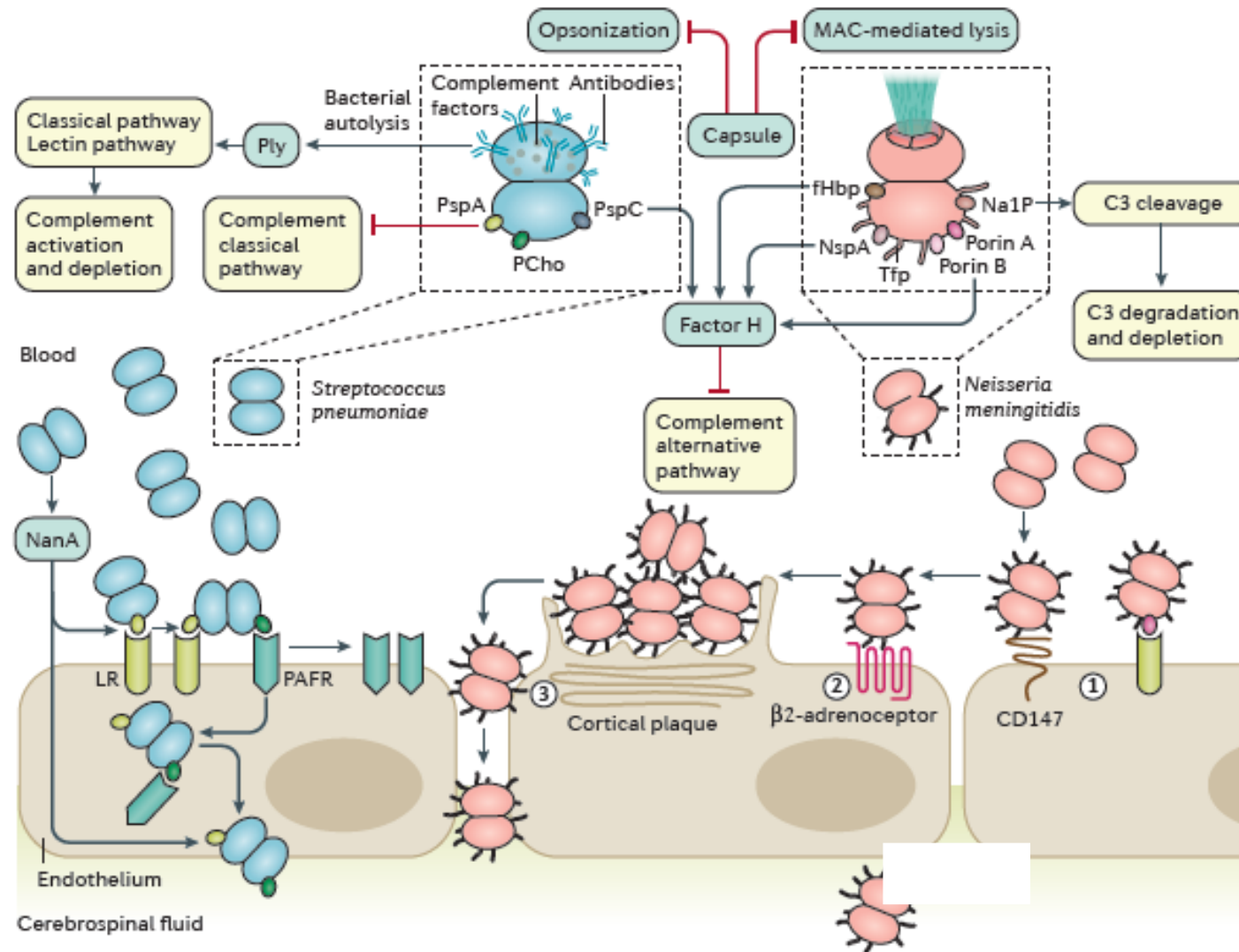


Figure 3 | Microbial survival in the bloodstream and central nervous system invasion. After initiation of

Patogenese

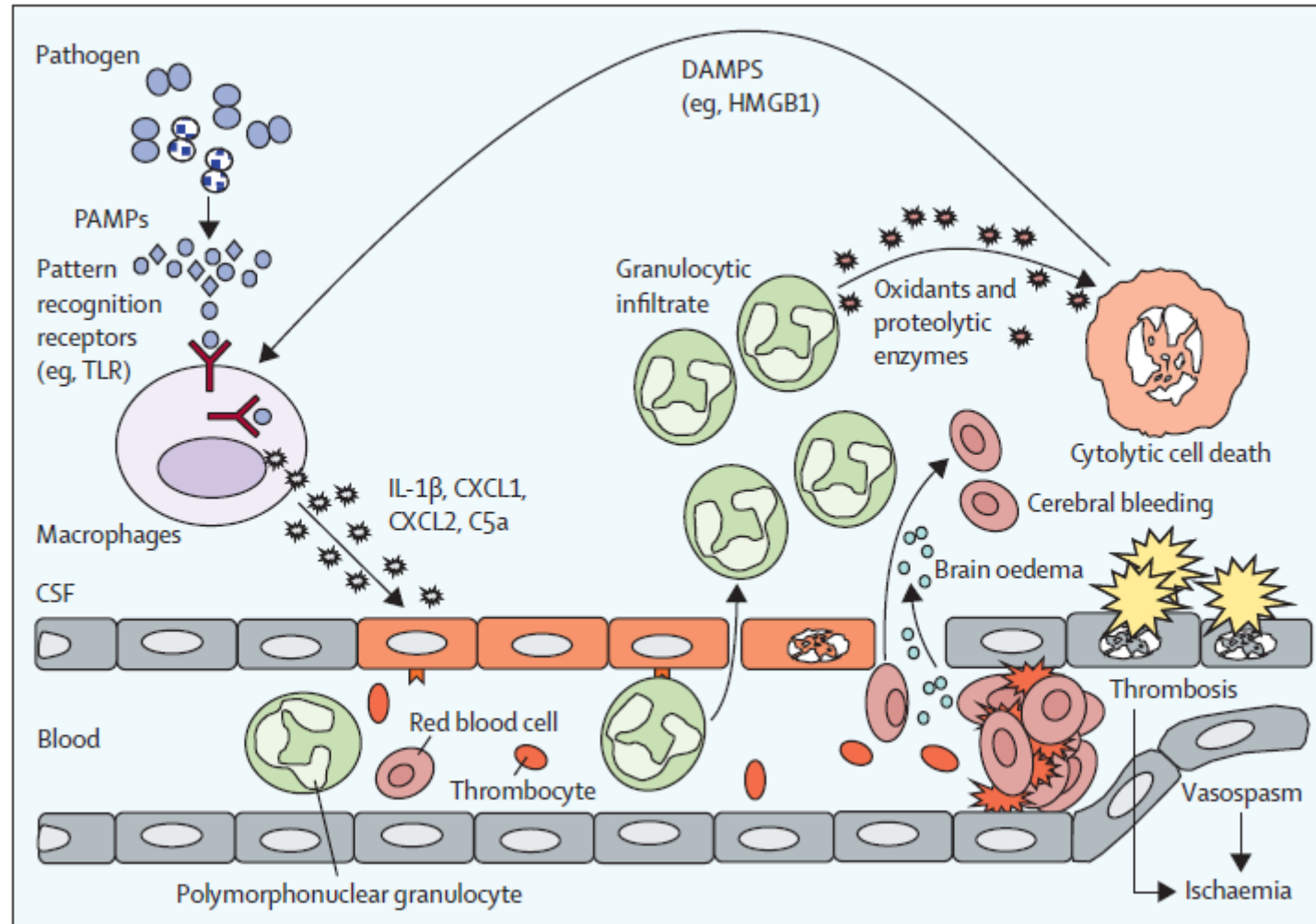
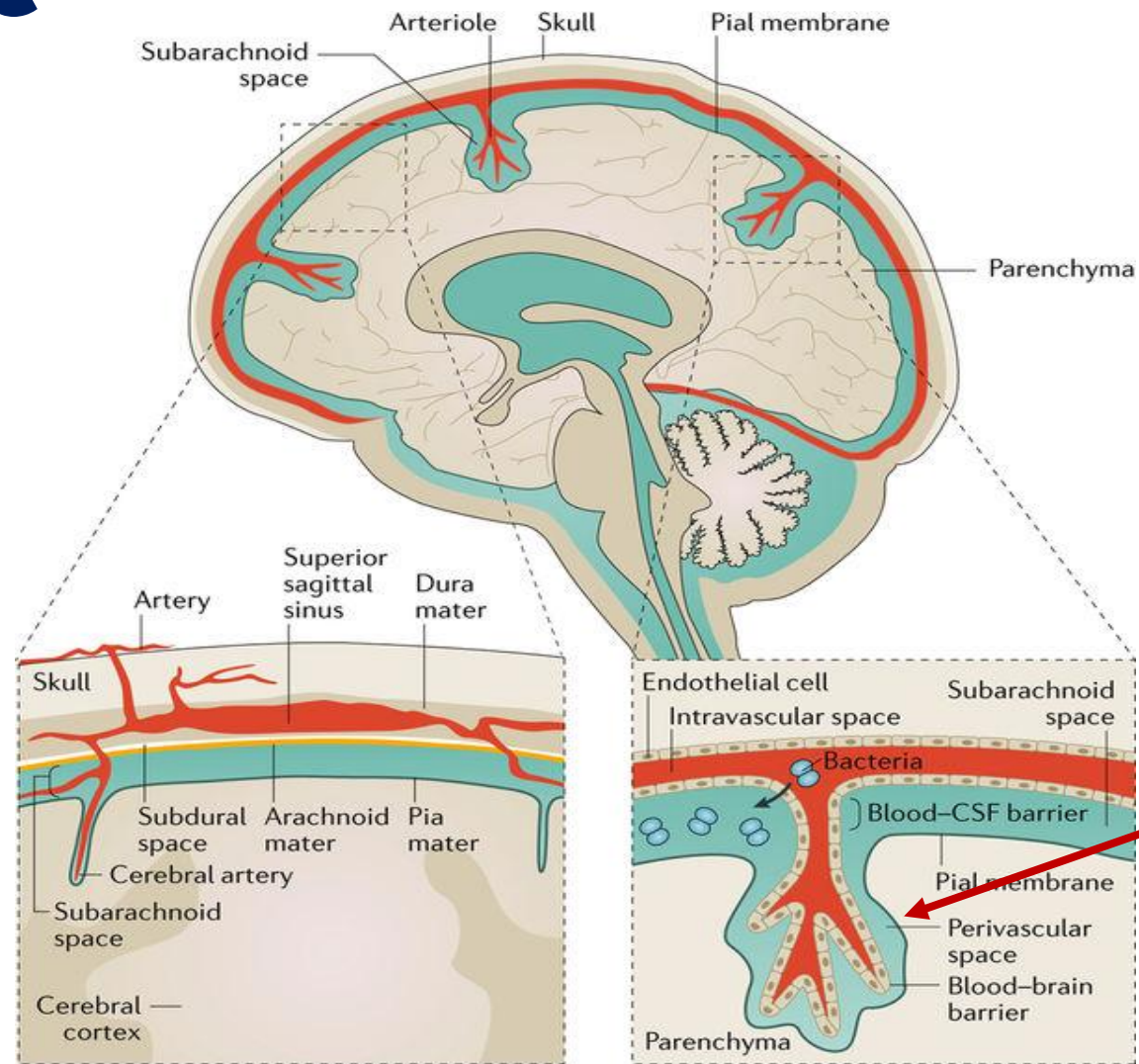


Figure 1: Schematic diagram of the key pathogenic steps of bacterial meningitis

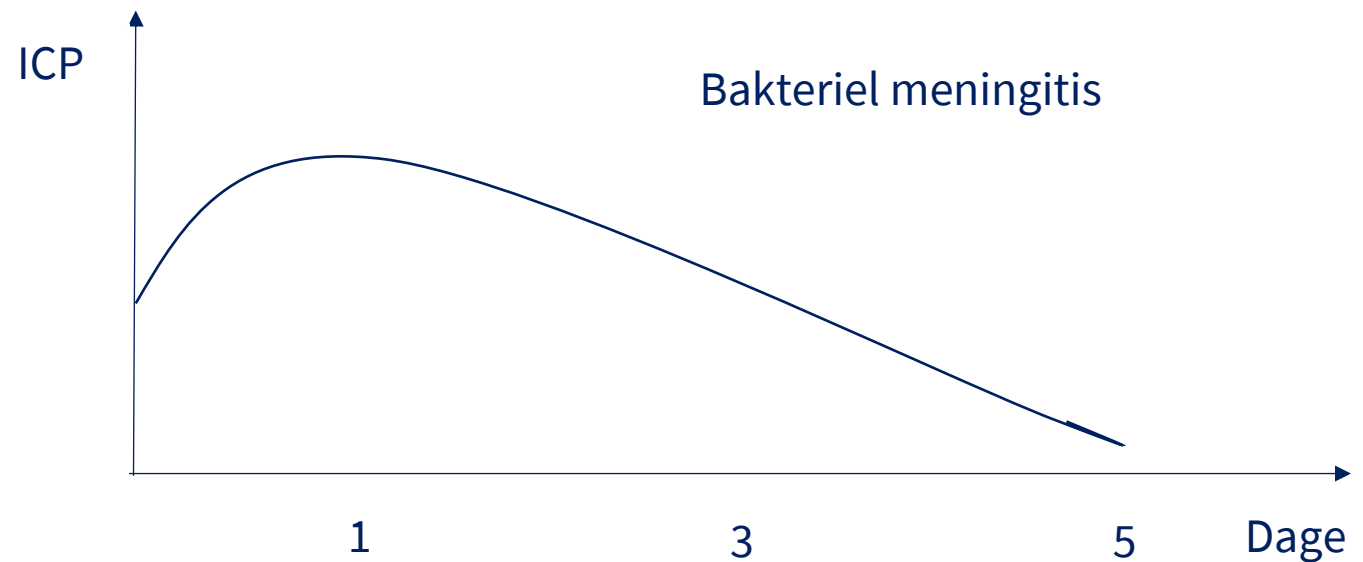
Patogenese

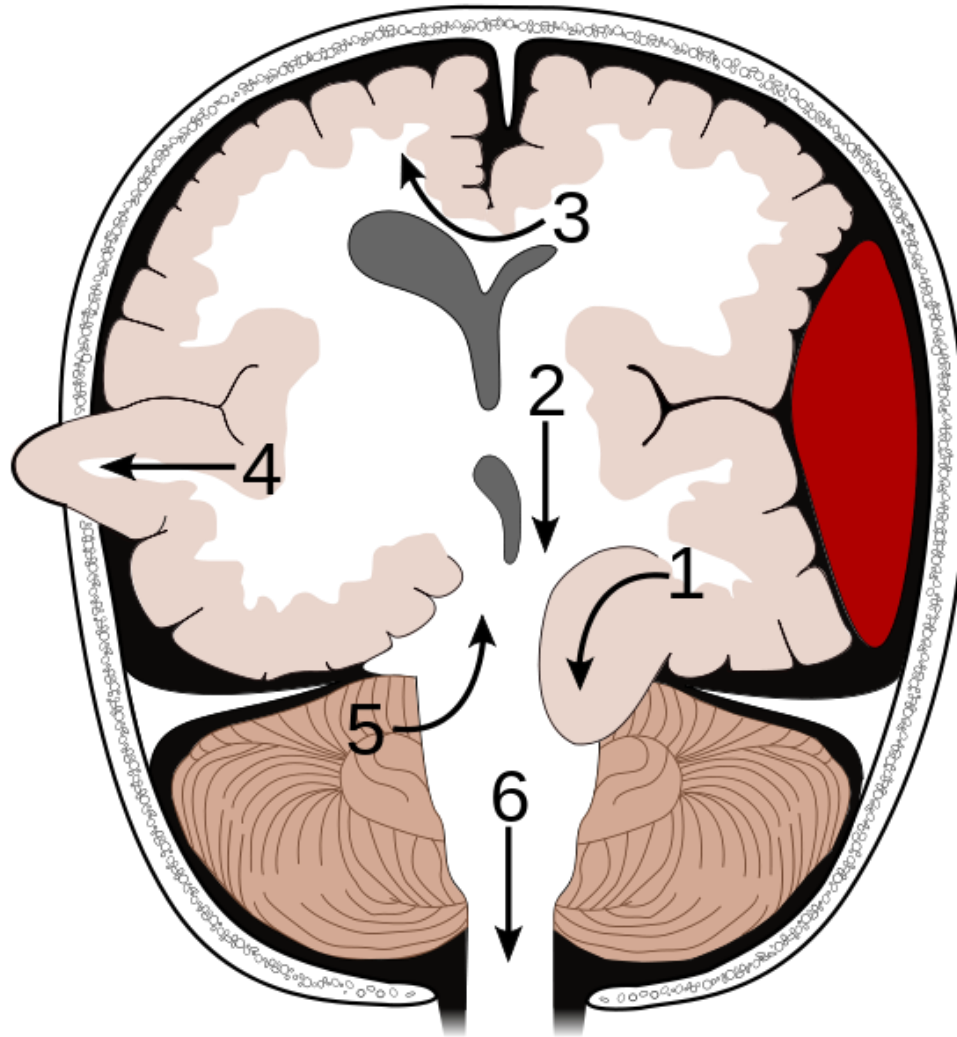


Virchow-Robin

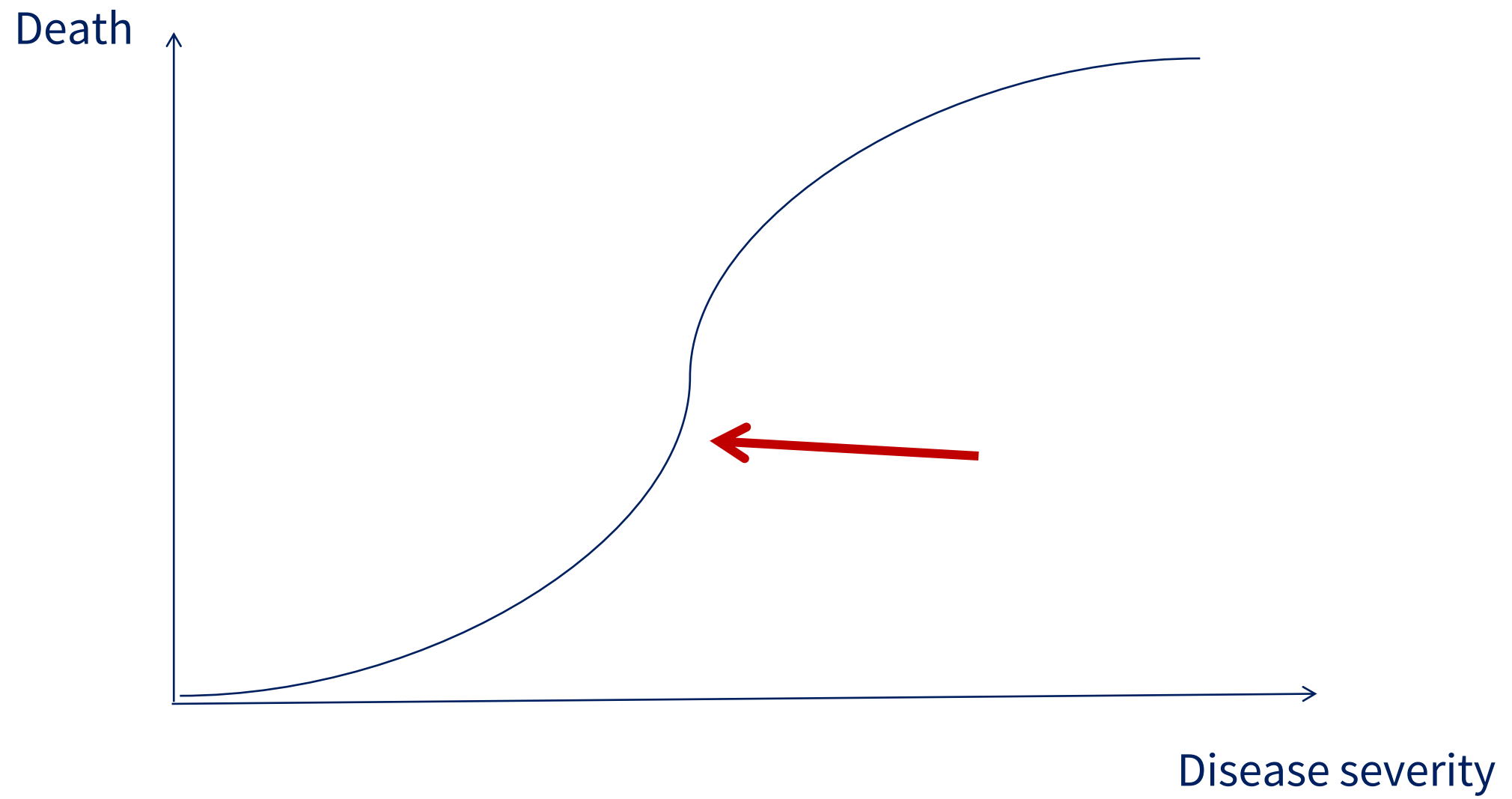
Forhøjet ICP ved bakteriel meningitis

- Inflammatorisk tilstopning af villi arachnoidales eller ventrikulære foramina
- Cerebralt ødem (vasogent, interstitielt, cytotoxisk)
- Sinus trombose



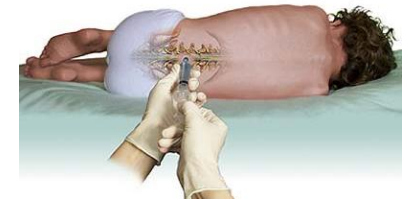


Meningitis: Inkarceration ved forhøjet CSF tryk **OG**
supra- eller infratentorial obstruktion!



Diagnose

- Anamnese
- Fysisk undersøgelse
- Bloddyrkning (80% pos) + biokemi
- LP og CSV undersøgelse



Klinisk præsentation

- Meningitis = meningeal inflammation
- Hyppige symptomer og fund

Triade (41%)	• Hovedpine	(83%)
	• Nakke- og rygstivhed	(74%)
	• Feber	(74%)
	• Ændret bevidsthedsniveau	(71%)

Differential diagnoser

- Enkelt center kohorte studie, n=363 OBS CNS infektion

Diagnose	No. (%)
CNS infektion	89 (24%)
Bakteriel meningitis	27 (7%)
Non-Infeksiøs CNS tilstand	119 (33%)
Anden infektion	111 (31%)
CNS inflammatoriske (autoimmune) sygdomme	36 (10%)
Andet	8 (2%)

10 hospitaler i England i 2010:
26/385 (7%) mistænkt for akut
CNS infektion havde encephalitis
eller bakteriel meningitis

Akutte CNS infektioner

	Bakteriel meningitis	Viral meningitis	Encephalitis
Hovedpine	++	+++	+(+)
NRS	++	+++	(+)
Feber	++	++	+
Ændret bevidsthedsniveau	+++	(+)	+++
Foto- eller fonofobi	+	++	-
Neurologisk udfald	+	-	+++
Systemisk påvirkning (sepsis)	+++	-	-
Symptomvarighed	1-2 dage	1-2 dage	1-5 dage

Prædiktive værdier

INGEN NRS, hovedpine, feber ELLER ændret bevidsthedsniveau: NPV 100% (95% CI 91-100)

Karakteristika (N=363*)	Sens	Spec	PPV	NPV
Hovedpine	63%	41%	7%	94%
Feber	38%	65%	7%	93%
Ændret bevidsthedsniveau	44%	71%	11%	94%
Nakke- og rygstivhed	63%	87%	28%	97%
Klassiske triade	26%	97%	44%	94%

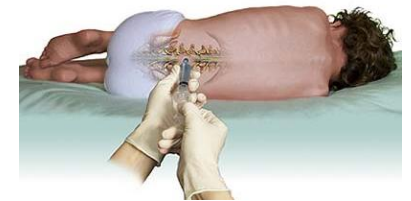
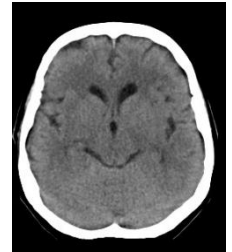
*27 med bakteriel meningitis

Diagnose

- Anamnese
- Fysisk undersøgelse
- Bloddyrkning (80% pos) + biokemi
- LP og CSV undersøgelse

**ALTID ved
mistanke**

**Kan ikke
udelukke CNS
infektion**



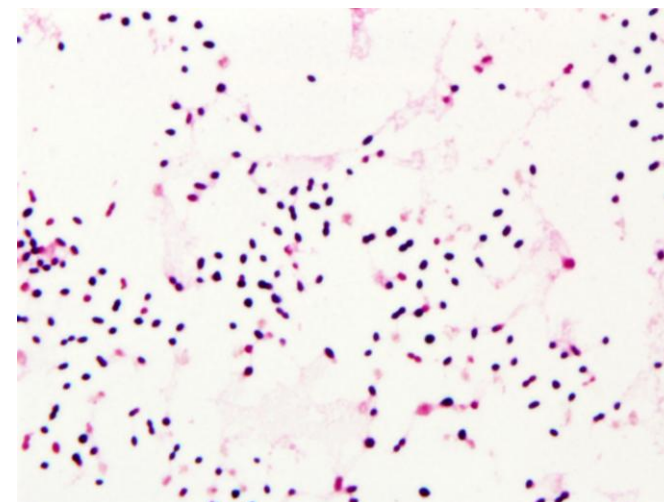
CSV biokemi

TYPISKE SPINALVÆSKEFUND VED FORSKELLIGE MENINGITISÅRSAGER							
	Åbnings-tryk (cm H ₂ O)	Leuko- cyttantal (celler/mL)	Dominerende celletype	Protein- koncen-tration (g/L)	Sp-glukose (mmol/L)	Sp-glukose/ P-glukose ratio	Sp-laktat (mmol/L)
Normal	12-20	< 5	-	< 0,5	2,3-3,9	> 0,6	< 3,5
Bakteriel (purulent)	Øget	Forhøjet, > 10 ² -10 ⁴	Neutrofile	Forhøjet	Lavt-meget lavt	Lavt-meget lavt	Forhøjet, > 3,5
Viral (serøs)	Normal-let øget	Forhøjet, 10 ² -10 ³	Lymfocytter	Let forhøjet	Normalt	Normalt	Normalt
Tuberkuløs	Øget	Forhøjet, 100-500	Lymfocytter	Meget forhøjet	Meget lavt	Meget lavt	Forhøjet
Svampe	Øget	Forhøjet, 100-500	Lymfocytter	Forhøjet	Lavt	Lavt	Forhøjet

CSV mikrobiologi

- Dyrkningspositive ≈50%
- Gram farvning
 - *S. pneumoniae* 70-90%
 - *N. meningitidis* 30-90%
 - Andre <80%
- PCR
 - *S. pneumoniae* 61-100%
 - *N. meningitidis* 88-94%
 - Andre ?

10% lavere risiko for
hver time efter AB



Obs risiko falsk positive
ved PCR panel

Særlige kendetegn...

Patogen	Karakteristika
Pneumokokker	Otitis media, sinuitis, pneumoni, triade
Meningokokker	Yngre, hududslæt, normalt bevidsthedsniveau
<i>L. monocytogenes</i>	Alder >50, immundefekt, længere symptomvarighed, variabelt CSV leukocytter
Haemophilus spp	Ældre, otitis media, sinuitis, basis cranii defekt
<i>S. aureus</i>	Endocarditis, osteomyelitis
Beta-haem strep	Ældre, endocarditis, alkoholmisbrug, otitis media, sinuitis
<i>E. coli</i>	Spædbørn og ældre, bakteriæmi
<i>C. canimorsus</i>	Hunde...

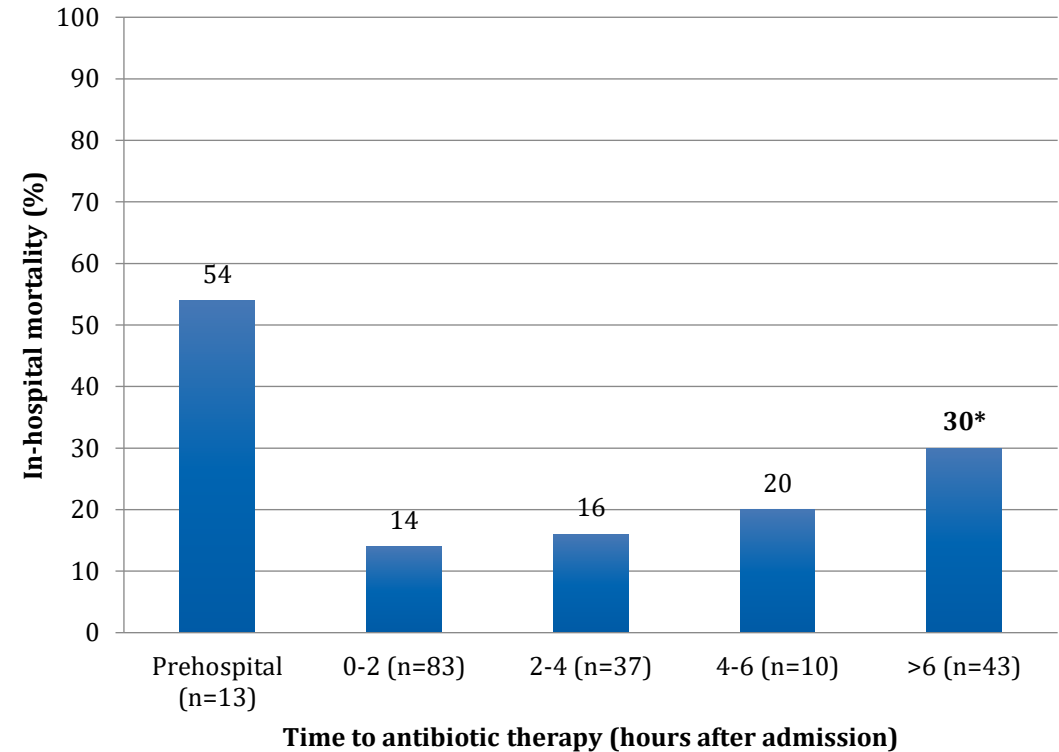
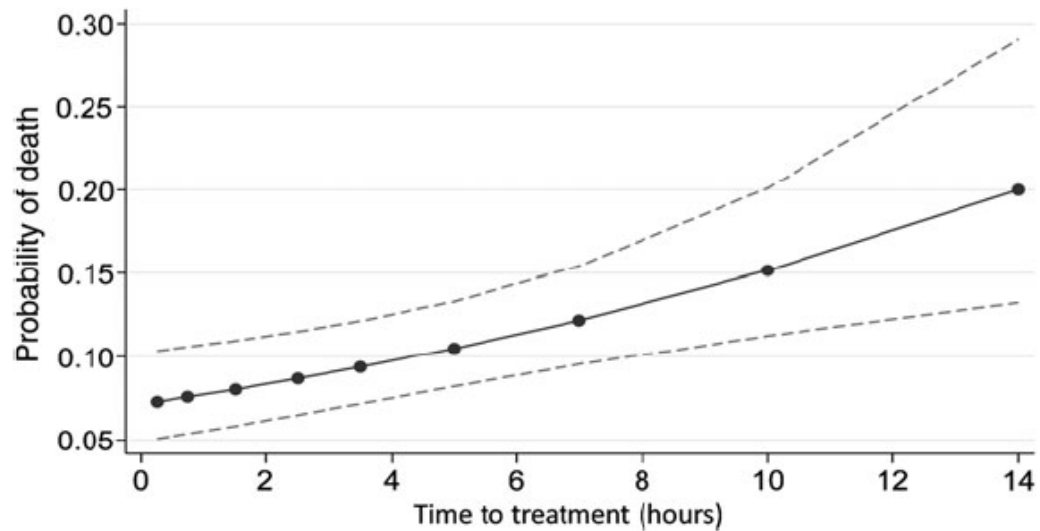
Kjærgaard, EJCMID, 2019; Søndermølle, in preparation, 2022; Bijlsma, Lancet Inf Dis, 2016; Heckenberg, Neurology, 2012; Koopmans, CID, 2016; Charlier, Lancet Infect Dis, 2017; Bodilsen, J Infect, 2018; Chekrouni, J Infect, 2021; Ter Horst, CID, 2021; Duerlund, in preparation, 2022

Udredning

- Bloddyrkning
- LP – **obs glas 1,2,3,4, flere...**
- Otoskopi
- Svælgpodning
- Evt. petekkie-podning
- Rtg thorax
- Evt. ekspektorat, urin
- Evt. PUT, MAT osv.



Tid til antibiotika og død



Behandling

- Dexamethason 10 mg x 4 i 4 dage
- Ceftriaxon 4 g x 1 + penicillin 3 MIE x 6
- Væske?
- Monitorering!

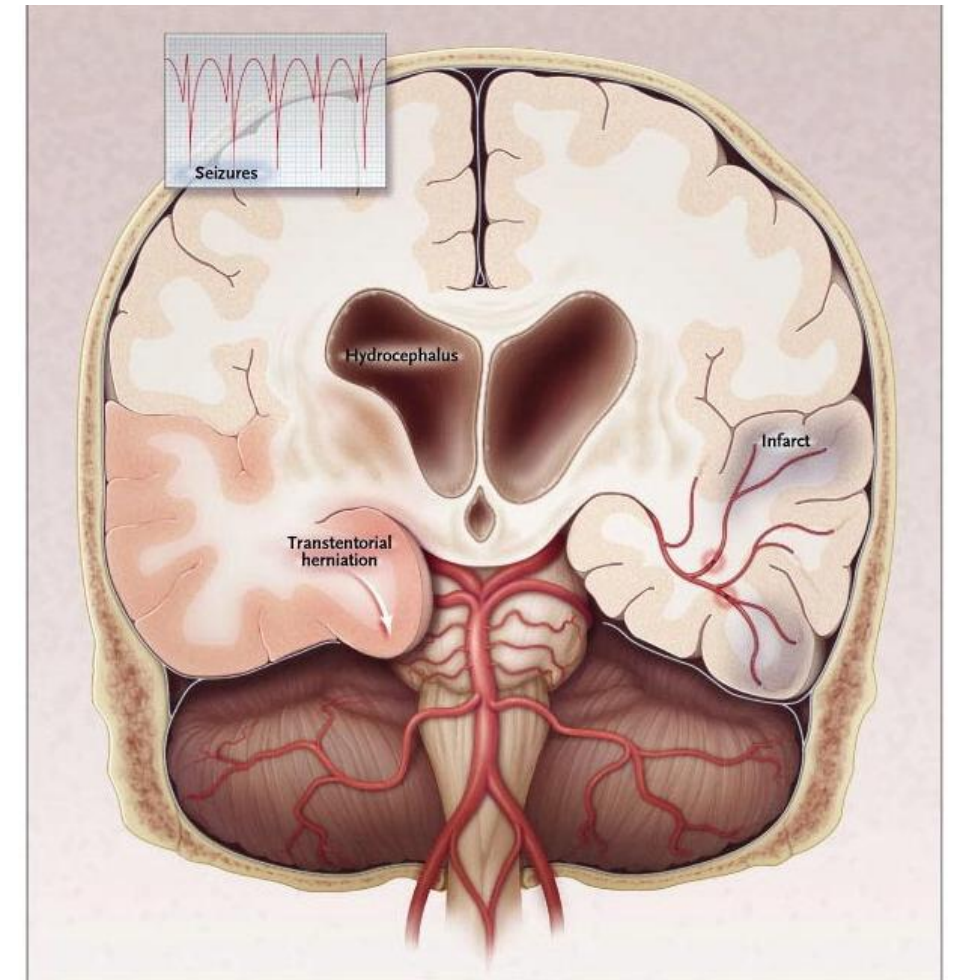
Hvorfor?

Normohydreret med mindre patienten er septisk



Komplikationer

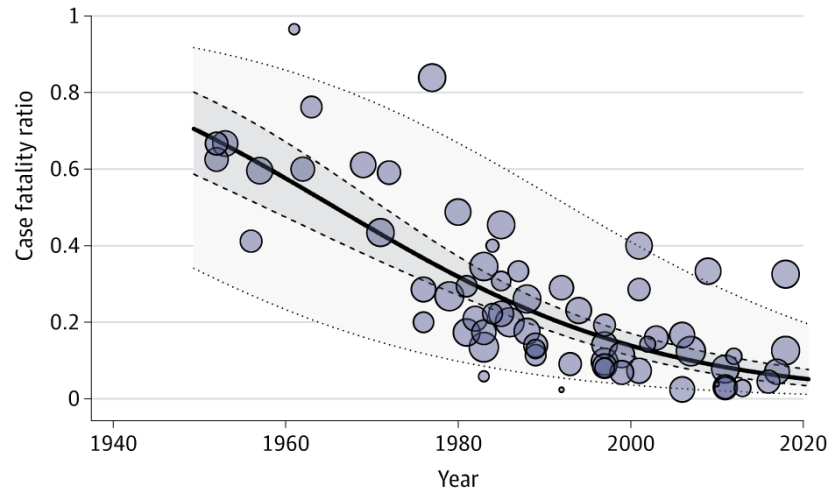
Komplikation	%
Neurologiske	60
Infarkt	25
Epilepsi	17
Hydrocephalus	5
Intrakraniel blødning	2
Subduralt empyem/absces	4
Høretab	22



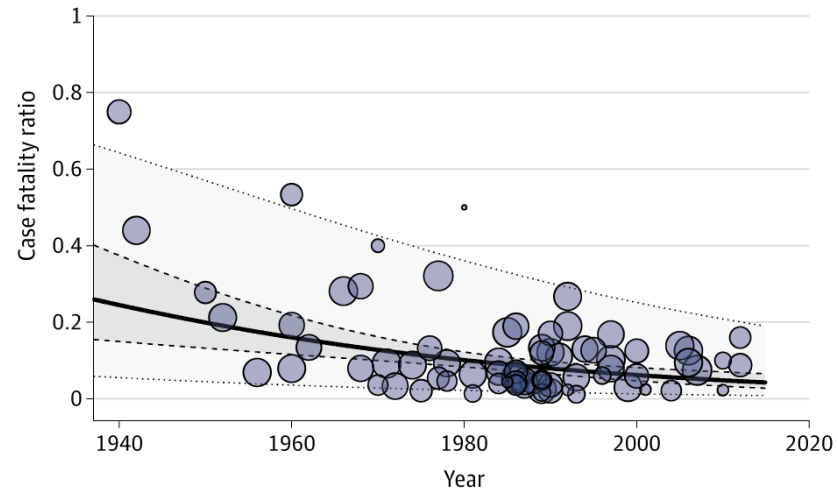
Van de Beek, NEJM, 2004; Costerus Lancet Neurol 2018;
Brouwer Lancet 2012; Brouwer, CMR, 2010

Case-fatality rates - overall

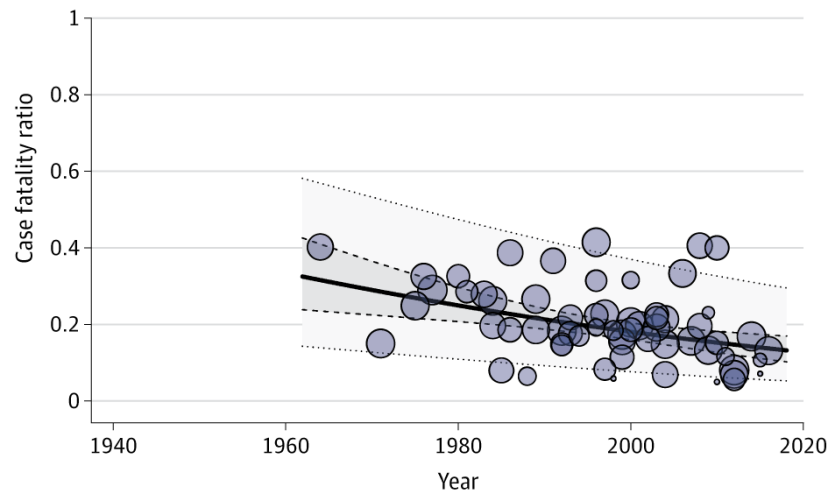
A Neonates



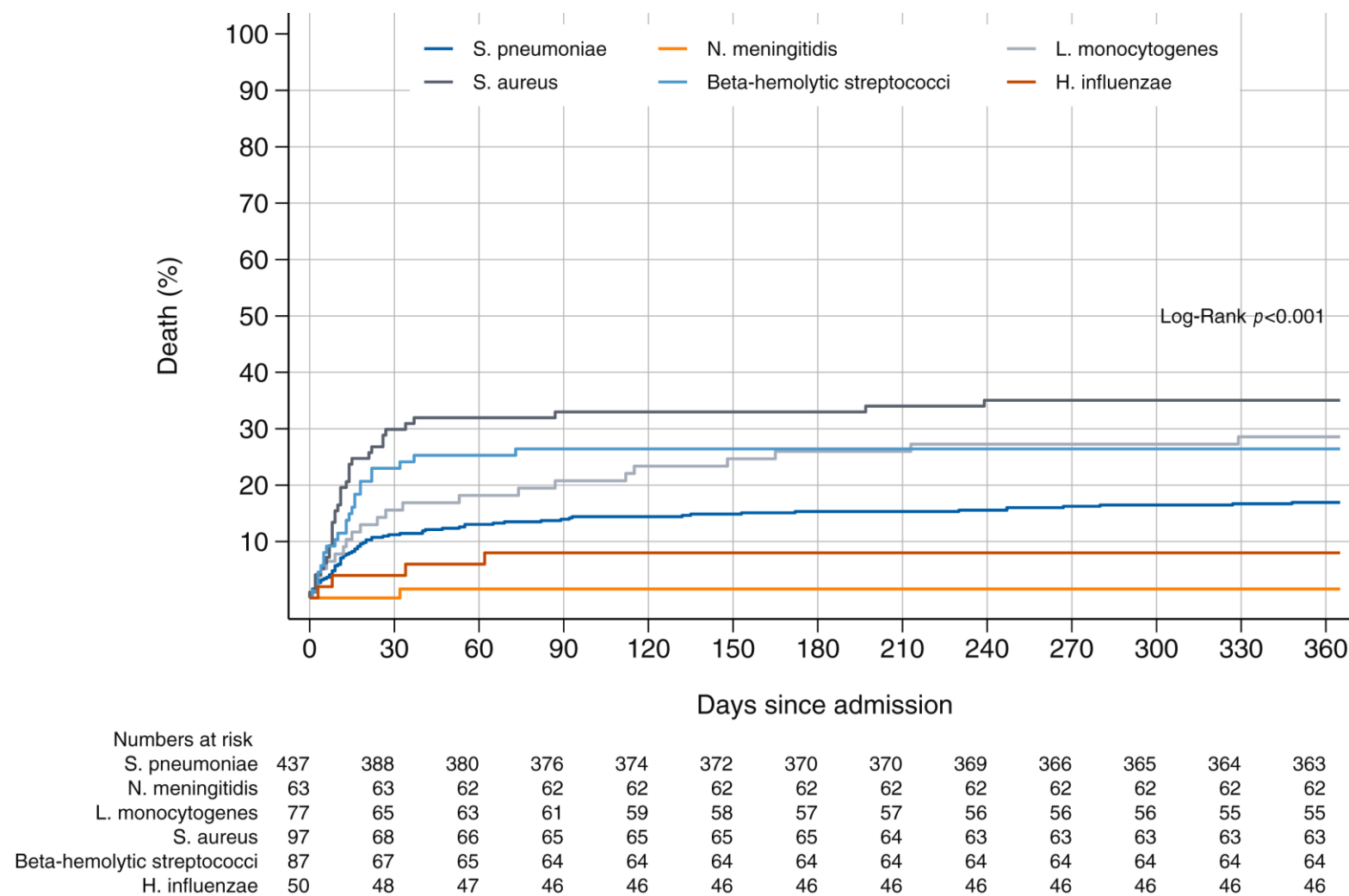
B Children



C Adults



Dødelighed i Dk



Future directions?

- Optimized corticosteroid treatment?
- More immunomodulation (C5a inhibitor)?
- Killing bacteria slowly?
- Hit 'em harder?
- ICP management?